MILITARY SPECIFICATION

METER, POWER FACTOR 4-1/2 INCH, SWITCHBOARD TYPE

1. SCOPE

1.1 This specification covers 4-1/2 inch rectangular, 250 degree scale, single and polyphase, shockproof power factor meters.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
LLL-B-631 - Boxes; Fiber Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid, (for Domestic Shipment).
PPP-B-601 - Boxes, Wood-Cleated-Plywood.

MILITARY

JAN-P-72 - Primer, Synthetic Lacquer-Resisting.
MIL-B-80 - Plastic Sheet, Acrylic Base, Antielectrostatic, Transparent, (for Indicator Dial Windows).
JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes, Wood-Cleated, Solid Fiberboard.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood Nailed.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.
MIL-P-116 - Preservation, Methods of.
MIL-C-490 - Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings.
JAN-P-735 - Primer, Paint, Zinc-Chromate Alkyd Type.
MIL-S-901 - Shockproof Equipment, Class H1 (High-Impact), Shipboard Application Tests for.
MIL-E-917 - Electric Power Equipment, Equipment for (Naval Shipboard Use).
MIL-S-[illegible]33 - Screws, Machine, Cap and Set; and Nuts.
MIL-I-[illegible]361 - Instruments, Electrical Measuring, Shunts, Resistors, and Transformers (Ship and Shore Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment (Naval Shipboard Use).
MIL-S-6872 - Soldering Process; General Specification for.
MIL-P-6889 - Primer, Zinc-Chromate, for Aircraft Use.
MIL-B-[illegible]0377 - Boxes, Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-[illegible]0547 - Liners, Case, Waterproof.

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MILITARY (cont'd.)
MIL-I-15024 - Identification plates, Information Plates and Marking Information for Identification of Electrical, Electronic, and Mechanical Equipment.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-C-15328 - Coating, Pretreatment, (Formula No. 117 for Metals).

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office Washington 25, D.C.)

3. REQUIREMENTS

3.1 Qualification.- Power factor meters furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.3).

3.2 Definitions.-

3.2.1 Power factor meter.- A power factor meter is an electrical instrument which visually indicates the present value of the relation between true power (watts) and apparent power (volt-amperes) and also indicates whether the current is lagging or leading the voltage.

3.2.2 Accuracy rating.- The accuracy rating designates the accuracy classification of the power factor meter. It is given as the limit, expressed as a pure number (decimal fraction) representing a percentage of a given length of scale which errors will not exceed when the meter is used under reference conditions.

3.2.3 Self-contained meter.- A self-contained meter is a meter which has all the necessary equipment built into the case or made a corporate part thereof.

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3.2.4 Mechanism.- The mechanism is the arrangement of parts for producing and controlling the motion of the indicating means. It includes all the essential parts necessary to produce these results, but does not include the base, cover, dial, or any parts, such as series resistors or shunts, whose function is to adapt the meter to the quantity to be measured.

3.2.5 Moving element.- The moving element of a meter comprises those parts which move as a direct result of a variation in the electrical quantity which the meter is measuring. The weight of the moving element includes one-half the weight of the springs.

3.2.6 Voltage circuit.- The voltage circuit is that combination of conductors and windings of the meter to which is applied the voltage of the circuit in which a given electrical quantity is to be measured, or a definite fraction of that voltage, or a voltage or current dependent upon it.

3.2.7 Current circuit.- The current circuit is that combination of conductors and windings of the meter which carries the current of the circuit in which a given electrical quantity is to be measured, or a definite fraction of that current, or a current dependent upon it.

3.2.8 Influence.- The influence of a specified variable or condition of a meter is the change in the indication of the meter caused solely by a departure of the specified variable from its reference value, all other variables being held constant.

3.2.9 Scale length.- The scale length of a meter is the length of the path described by the indicating means or the tip of the pointer in moving from one end of the scale to the other. In the case of knife-edge pointers and others extending beyond the scale division marks, the pointer shall be considered as ending at the outer end of the shortest scale division marks.

3.2.10 Damping.- Damping of a meter is the term applied to its performance to denote the manner in which the pointer settles to its steady indication after a change in the value of the measured quantity. Two general classes of damped motion are distinguished as follows:

(a) Periodic, in which the pointer oscillates about the final position before coming to rest.
(b) Aperiodic, in which the pointer comes to rest without overshooting the rest position.

3.2.11 Period.- The period is the time between two consecutive transits of the pointer or indicating means in the same direction through the rest period.

3.2.12 Overshoot.- Overshoot is the ratio of the overtravel of the indicator beyond its steady deflection to the change in steady deflection when the instrument is suddenly energized with a new constant value of the measured quantity. The overtravel and deflection are determined in angular measure and the overshoot is usually expressed as a percentage. When the ratio depends on the magnitude of the deflection, a value corresponding to an initial deflection from zero to end scale is used in determining the overshoot for rating purposes.

3.2.13 Damping factor.- The damping factor is the ratio of the deviations of the pointer in two consecutive swings from the position of equilibrium, the greater deviation being divided by the lesser. The deviations are expressed in angular measure. When the damping factor depends upon the magnitude of the deflection, it is measured as the ratio in angular degrees of the steady deflection to the difference between angular maximum momentary deflection and angular steady deflection produced by a sudden application of a constant electric power. The damping factor shall be determined with sufficient constant electric power applied to carry the pointer to full scale deflection on the first swing. The damping shall be due to the meter and its normal accessories only.

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3.2.13.1 In determining the damping factor, the rated voltage and current shall be applied with such phase relation as to give center scale deflection. The pointer shall then be moved mechanically to end scale and suddenly released.

3.2.14 Response time.- The response time is the time required after an abrupt change has occurred in the measured quantity to a new constant value until the pointer, or indicating means, has first come to apparent rest in its new position. When the response time depends on the magnitude of the deflection, a value corresponding to an initial deflection from zero to end scale is used in determining the response time for rating purposes. When this is the case, the pointer is at apparent rest when it remains within a range on either side of its final position equal to one-half the accuracy rating. In determining the response time the rated voltage and current shall be applied with such phase relation as to give center scale deflection. The pointer shall then be moved mechanically to end scale and suddenly released.

3.2.15 Current loss.- The current loss is the current in the voltage circuit at rated voltage.

3.2.16 Voltage loss.- The voltage loss is the value of the voltage between the terminals at rated current.

3.2.17 Rated voltage.- The rated voltage is the normal continuous operating voltage and is the normal value of applied voltage used for test purposes.

3.2.18 Maximum voltage.- The maximum voltage is a stated voltage, which, if applied continuously at maximum stated operating temperature and with any other circuits in the meter energized at rated values, will not cause electrical breakdown or any observable physical degradation.

3.2.19 Rated current.- The rated current is the normal value of current used for design purposes.

3.2.20 Maximum current.- The maximum current is a stated current which, if applied continuously at maximum stated operating temperature and with any other circuits in the instrument energized at rated values, will not cause electrical breakdown or any observable physical degradation.

3.3 Material.- The material for each part shall be as specified herein. In general, material requirements shall be the applicable requirements of Specification MIL-E-[illegible] unless otherwise specified herein. When a definite material is not specified, a material shall be used which is adequate for the purpose intended and for meeting the requirements herein. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4 Design and construction.-

3.4.1 Balancing.- The moving element shall be balanced by a means which will provide an easy method of rebalancing, such as threaded lock- or split-nuts, or a wire helix moving along a cross arm and counterweight arm. The use of solder or an adhesive such as shellac, to hold the balance weights or to function as a part of the balance weight, will not be acceptable.

3.4.2 Magnets.- If laminated or multiple-piece magnets are used, the laminations or components shall be securely fastened together so that they will not separate or loosen as a result of tests specified herein.

3.4.3 Threaded parts.- All threaded parts shall be in accordance with Handbook H28.

3.4.3.1 Bolts, nuts, and machine screws.- All bolts, nuts and machine screws shall conform to Specification MIL-S-[illegible].

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3.4.3.2 Locking of screw-thread assemblies.- All screw-thread assemblies shall be made vibrationproof. When practicable, split-type lock washers or equivalent means shall be provided under all nuts.

3.5 Design and construction of component parts.-

3.5.1 Cases.- Meter cases shall be made of molded thermosetting plastic material or metal. The case shall be of dripproof construction in accordance with Specification MIL-E-2036 and Standard MIL-STD-108. The dimensions of the meter case shall be as specified herein.

3.5.2 Finish.- The exposed metal parts of the power factor meter case shall be finished in accordance with the following.

3.5.2.1 Cleaning.- After all fabricating operations have been completed, all corrosive products, grease, oil, dirt or any other foreign matter shall be removed by solvent wiping, vapor degreasing, or caustic washing and rinsing. This shall be done in accordance with the best commercial practices.

3.5.2.2 Pretreatment.- After cleaning, the pretreatment shall be accomplished in accordance with either of the following two methods:

(a) One coat of primer pretreatment coating in accordance with Specification MIL-C-15328, applied as a continuous film 0.0002 to 0.0005 inch thick.
(b) A phosphate treatment conforming to grade I of Specification MIL-C-490.

3.5.2.3 Primer.- Following the pretreatment specified, one coat of primer in accordance with Specification JAN-P-72, JAN-P-735, or MIL-P-6889 shall be applied as a continuous film, 0.0004 to 0.001 inch thick.

3.5.2.4 Enamel.- After the primer, one coat of gray enamel conforming to either type II or type III, class 2 of Specification MIL-E-15090 shall be applied as a continuous film, approximately 0.001 inch thick.

3.5.3 Scale.- The minimum scale length shall be 5 inches. The scale shall be marked from zero to unity power factor lagging and from zero to unity power factor leading. Unity power factor shall be at the center and top of the scale.

3.5.4 Window.- Meters shall be provided with a window of glass, or other transparent material, free from discoloration, scratches, striae, electrostatic effect or other detrimental defects which may interfere in any way with normal reading of the meter in daylight. Plastic windows, where used, shall be in accordance with Specification MIL-P-80. The window shall be secured by clamps, or by other suitable means. Cement alone is not considered sufficient for holding the window in place.

3.5.5 Scale visibility.- The scale and window shall be so designed that when the dial is vertical the entire scale of the meter can be easily read at any angle up to 45 degrees below the horizontal centerline or the vertical centerline, and 20 degrees to either side of the vertical centerline on the horizontal centerline, with general illumination coming from above and in front and at an angle 45 degrees from the horizontal centerline. Under these conditions there shall be no objectionable shadow on the graduated scale.

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3.5.6 Dial and pointer.-

3.5.6.1 Dial.- The dial shall be made of stiff material firmly secured to its mounting and readily removable. The dial shall be of such material and so secured as to insure freedom from warping, fading, or other deterioration during the tests specified herein. Paper dials will be satisfactory only when adequately bonded to and backed by a metal support and when they meet all tests specified herein without fading, discoloring, warping, or other deterioration.

3.5.6.2 Pointer.- The meter shall have a pointer so formed as to permit accurate readings at the usual distance from which the meter is viewed, and approximate readings up to the distance of legibility of the scale markings. The pointer shall be of rigid construction and suitable for the intended application, and shall not become damaged or distorted when subjected to any of the tests specified herein.

3.5.6.2.1 Pointer clearance (parallax).- Where the meter is not provided with means for avoiding errors in reading due to parallax, the clearance between the pointer index and the graduated scale shall not exceed 0.10 inch, nor be less than 0.03 inch.

3.5.7 Means of adjustment.-

3.5.7.1 Accuracy adjustment.- Meters shall be provided with practical shop means of adjustment to accuracy. Charging of the magnet and magnetically treating it to obtain accuracy, or adjusting the internal resistor, shall be considered as meeting this requirement.

3.5.8 Terminal studs.- Terminal studs shall be firmly anchored to the base of the meter in such a manner as to prevent turning or movement sufficient to cause damage to internal construction. Each terminal stud shall be furnished with the necessary nuts or screws, and washers.

3.5.9 Connection diagrams.- Meters having more than one circuit shall have a print showing the wiring diagram of connections with or without transformers, resistances, and reactances for three-phase three-wire systems and two-phase four-wire systems, together with instructions attached permanently or sufficiently secured to prevent loss during shipment.

3.5.10 Identification plates.- The meter shall be provided with an identification plate conforming to Specification MIL-I-15024. All meter rating information shall be indicated on the identification plate. Plastic identification plates shall not be used when the plate is to be attached to other than a flat surface.

3.5.11 Self-contained limitations.- Power factor meters shall be self-contained a current ratings of 20 amps or less. External accessories may be used for all potential ranges.

3.5.12 Mounting.- Switchboard meters shall be mounted by means of mounting bolts or studs. The necessary mounting bolts (screws), nuts, and lock washers shall be supplied with each meter. The meters shall be mounted on a panel drilled in accordance with the panel-drilling specified (see 3.5.14).

3.5.13 External accessories.- External shunts, shunt leads, resistors, and transformers shall be furnished when specified (see 6.2). External accessories shall conform to Specification MIL-I-.361.

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3.5.14 Dimensions and drilling shall be as shown on figure 1.-

3.5.15 Soldering.- Soldering shall be in accordance with Specification MIL-S-6872.

3.6 Performance.- Power factor meters shall be designed and constructed to perform in accordance with the following requirements:

3.6.1 Calibration.- Power factor meters shall be calibrated at reference conditions as follows:

Position	Vertical.
Temperature	25° Centigrade (C.).
Mounting	Nonmagnetic panel and magnetic-steel panel 1/8 inch thick.
Frequency	60 cycles.
Phase	Single or polyphase.

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3.6.2 Performance ratings.- Power factor meters shall be designed and constructed so as to meet the following performance and environmental conditions and limitations:

Item	Requirement	Test reference paragraph
Rated accuracy	Within 0.02 from 20 to 40 percent, inclusive, rated current, and within 0.01 from more than 40 up to and including 120 percent rated current for a scale length of 0.5 lagging to 0.5 leading, inclusive, with rated voltage and frequency applied.	4.6.3
Position influence	60-degree tilt from normal operating position, 1 percent of scale length maximum with rated voltage and current applied.	4.6.2
Damping factor	3 minimum at rated voltage and 2/3 rated current (overshoot, 33 percent maximum).	4.6.4
Response time	2.5 seconds maximum at rated voltage and 2/3 rated current.	4.6.5
Power consumption	Potential system 7.5 volt amperes maximum per phase at 120 volts for nominal 120-volt rating. For higher nominal voltage ratings the maximum volt amperes shall be increased proportionately. Current system 4.5 volt amperes maximum at 60 cycles.	4.6.6
External temperature influence	0.01 maximum per plus or minus 10° C. variation from plus 25° C. with rated voltage and 2/3 rated current applied, temporary effect. 0.01 maximum permanent effect.	4.6.9
External field influence	0.03 maximum in 5 oersted field with rated voltage and 2/3 rated current applied.	4.6.8
Frequency influence	60 cycle polyphase meters 0.015 with ±10 percent deviation from rated frequency, with rated voltage and 2/3 rated current applied. 60 cycle single phase meter 0.015 maximum at any frequency from 59 to 61 cycles per second, inclusive.	4.6.7
Momentary overload	Rated accuracy.[1]	4.6.10.1
Sustained overload	Temporary - rated accuracy.[1] Permanent - 1/2 rated accuracy.[1]	4.6.10.2

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Item	Requirement	Test reference paragraph
Insulation resistance	1100 volts maximum, peak value.	4.6.13
Dielectric test	Live parts to face and panel: 2600 volts a.c. between any independent circuits: 1500 volts a.c.	4.6.12
Maximum voltage	20 percent greater than rated value.	-----
Maximum current	20 percent greater than rated value.	-----
Voltage influence	0.006 maximum with plus or minus 10 percent voltage variation from nominal voltage rating: 4 times this maximum error is allowed for a plus or minus 20 percent variation.	4.6.11
Humidity resistance	Twice rated accuracy limits.[1]	4.6.16
Vibration	Rated accuracy limits.[1]	4.6.14
Friction effect	Twice rated accuracy.[1]	4.6.15
Shock	Three times rated accuracy limits.[1]	4.6.17

[1]This deviation is in addition to the rated accuracy deviation. All deviation values in terms of unity.

3.7 Workmanship.- Meters shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and sound practice. The interiors of meters shall be free from metal filings, grease or oil, foreign material, dust, or other loose particles which might interfere with the normal operation of the meters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- The quality assurance tests conducted on power factor meters shall be classified as follows:

(a) Qualification inspection.
(b) Acceptance inspection.

4.2 Test conditions and general instructions.-

4.2.1 Atmospheric conditions.-

4.2.1.1 Qualification tests.- Unless otherwise specified herein, all qualification tests shall be made at ambient temperature of 23° to 27° C. at existing barometric pressure, and with relative humidity of 40 to 60 percent.

4.2.1.2 Inspection tests.- Unless otherwise specified herein, all inspection tests shall be made at ambient temperature of 20° to 30° C. at existing barometric pressure, and with relative humidity less than 80 percent.

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4.2.2 Reference temperature.- The reference temperature shall be 25° C.

4.2.3 Position.- Unless otherwise specified herein, tests performed on meters, including all tests requiring scale readings, shall be performed with the meter in the position in which it is normally used.

4.2.4 Examination of pivots and bearings.- Examination of pivots and bearings for evidence of corrosion, damage from vibration shock and other tests, shall be made through a microscope having a magnification of 40 to 100 diameters.

4.2.5 Tapping.- When taking meter readings, either the meter or its support shall be tapped lightly, except when ascertaining the effect of friction after the vibration test.

4.2.6 Parallax.- Care should be exercised to avoid the effect of parallax when taking meter readings.

4.2.7 Thermal stability.- For all tests, care shall be taken to obtain thermal stability at the specified temperature prior to reading the meter.

4.2.8 Standard meters.- Standard meters shall be used to check the indications of meters submitted for test. Accuracies and tolerances specified are absolute, and readings of standard meters shall be corrected before determining compliance.

4.3 Qualification inspection at a Government laboratory.- Qualification inspection shall be conducted at a Government laboratory designated by the Bureau of Ships. This inspection shall consist of the tests specified in 4.6.

4.3.1 Samples.- Sample meters submitted for qualification approval shall be representative of the manufacturer's normal production and shall be taken, preferably, from a current production lot. Two specimens of each type of power factor meter (single phase or polyphase) for which qualification is sought shall be submitted for qualification inspection. Sample meters which are used with external shunts shall be supplied with shunt leads of [illegible] resistance for switchboard meters.

4.3.2 Failures.- One failure will be permitted on a sample of power factor meters under the following conditions.

(a) The failure is limited to one of the following tests:
  (1) Position influence (balance).
  (2) Rated accuracy.
  (3) Damping factor.
  (4) Response time.
(b) The magnitude of the failure is limited to 1-1/2 times the specified tolerance for the rated-accuracy test, and 2 times the specified tolerance for the other three tests.

4.4 Inspection.- Inspection shall be performed at the place of manufacture by the Government or under the direction of the Government.

4.5 Lot.- All power factor meters of the same design offered for delivery at one time shall be considered a lot.

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4.4.2 Sampling for group A tests.- Statistical sampling for group A tests shall be in accordance with Standard MIL-STD-105. Major and minor defects shall be as defined in Standard MIL-STD-105 and the Acceptable Quality Level (AQL) shall be as specified in 4.5.1.

4.4.3 Sampling for group B tests.- Group B tests shall be performed on sample units that have been subjected to and have passed the group A tests, unless the Government considers it more practical to select a separate sample from the lot for group B tests. The sampling procedure shall be in accordance with Inspection Level L7 of appendix to Standard MIL-STD-105. The Acceptable Quality Level (AQL) shall be 6.5 percent.

4.4.4 Sampling for group C tests.- One sample power factor meter from each lot regardless of the quantity shall be selected for the group C tests.

4.5 Lot acceptance inspection.-

4.5.1 Group A tests.- Each of the sample power factor meters selected in accordance with 4.4.2 shall be subjected to each of the tests specified in table I.

Table I - Group A tests.

Test	Requirement paragraph	Test paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical inspection (without opening case)	Section 3	4.6.1	1.0	4.0
Position influence (balances)	3.6.2	4.6.2	1.0	---
Rated accuracy	3.6.2	4.6.3		

4.5.2 Group B tests - Samples selected in accordance with 4.4.3 shall be subjected to each of the tests specified in table II. Failure to conform to this specification for any group B test shall be counted as a defect. If the number of defective units exceeds the applicable acceptance number, the lot shall be rejected.

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4.5.2.1 If a lot is rejected due to failure to pass group B tests, the manufacturer may rework the lot or screen out defectives and resubmit it for Government inspection. Resubmittal lots shall be kept separate from new lots. The resubmitted lot shall be inspected, using inspection level L7 of appendix to Standard MIL-STD-105. Sample units which have been subjected to and have passed the group B tests shall be subjected again to the rated accuracy test.

Table II - Group B tests.

Test	Requirement paragraph	Test paragraph
Rated accuracy	3.6.2	4.6.3
Damping factor	3.6.2	4.6.4
Response time	3.6.2	4.6.5
Power consumption	3.6.2	4.6.6
Frequency influence	3.6.2	4.6.7
Voltage influence	3.6.2	4.6.11
Dielectric stress	3.6.2	4.6.12
Insulation resistance	3.6.2	4.6.13

4.5.3 Group C tests.- Samples selected in accordance with 4 4 4 shall be subjected to the group C tests specified in table III. Any defective units shall be cause for rejection.

4.5.3.1 If a sample fails group C tests, the manufacturer shall take corrective action on the processes and all units of product which were manufactured with the same conditions, materials, processes, and are considered subject to the same test failure. Government inspection may be discontinued, at the option of the Government inspector, until a sample produced by the corrected process has been subjected to and has passed the group C tests failed by the preceding sample. Units of product subsequently submitted for Government inspection shall incorporate the corrective action, unless this requirement is waived temporarily by the bureau or agency concerned.

Table III - Group C tests.

Test	Requirement paragraph	Test paragraph
External field influence	3.6.2	4.6.8
External temperature influence	3.6.2	4.6.9
Overload	3.6.2	4.6.10

4.6 Test procedures.-

4.6.1 Visual and mechanical inspection.- Meters shall be inspected to determine compliance with all requirements of this specification which do not involve tests.

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4.6.2 Position influence.- The pointer shall be energized so that the pointer is set at any convenient position on the scale between 0.6 and 1.0 power factor, both lag and lead with the meter in the normal operating position. The power factor meter shall then be tilted 60 degrees from these positions in any direction and the deviation of the pointer noted (see 3.6.2).

4.6.3 Rated accuracy.- Power factor meters shall be checked for accuracy as specified in 4.6.3.1 and 4.6.3.2 as applicable.

4.6.3.1 Qualification inspection.- The qualification test for rated accuracy shall be performed at any four convenient points on the scale, two between 0.1 and 1.0 power factor lagging two between 0.1 and 1.0 power factor leading. The meter shall be conditioned in the test circuit for 30 minutes for each of the 4 points prior to readings. Readings shall be taken and compared with corresponding readings of the standard meter (see 3.6.2).

4.6.3.2 Acceptance inspection.- The acceptance test for rated accuracy shall be performed under the test conditions of 4.6.3.1 except that only one lag and one lead reading is necessary and preliminary conditioning is not required.

4.6.4 Damping factor.- Electric power sufficient to produce momentary end-scale deflection shall be applied to the meter. The maximum momentary deflection shall be observed and recorded. After the pointer has come to rest, the steady deflection shall be observed and recorded. When making this test, the circuit resistance shall be not less than 100,000 ohms and such that no additional damping is added to the system. The damping factor shall be taken as the ratio of the steady deflection (in angular units) to the difference between the maximum momentary deflection and the steady deflection in angular units. Example: If the maximum momentary deflection is 90 angular degrees and the steady deflection is 75 angular degrees, the difference between the two is 15 degrees, and the damping factor (the ratio of 75 degrees to 15 degrees) is 5 (see 3.6.2).

4.6.5 Response time.- Steady electric power sufficient to produce momentary end-scale deflection shall be applied to the meter. The length of time, in seconds, required for the pointer to come to apparent rest shall be recorded. The pointer shall be considered as having come to apparent rest when it has obtained the actual point of rest within one-half the initial accuracy requirement. The test shall be repeated five times, and the average length of time recorded for all measurements shall be considered as the response time of the meter (see 3.6.2).

4.6.6 Power consumption.- The power consumption shall be measured at those points on the scale (both lag and lead) which will produce the maximum values (see 3.6.2).

4.6.7 Frequency influence.- Power factor meters shall be calibrated at nominal frequency. Polyphase meters then shall be subjected to frequencies 10 percent above and 10 percent below the nominal frequency under conditions specified in 3.6.2. Single phase meters shall be subjected to plus or minus one cycle deviation of nominal frequency under conditions specified in 3.6.2.

4.6.8 External-field influence.- The meter shall be placed at the center of a four-turn coil, 30 centimeters in diameter, carrying a current of 50 amperes of the same kind and frequency as that which actuates the meter mechanism, and which produces a field of approximately 5 oersteds at the center of the coil. Readings shall be taken with the meter at various positions in the center of the field at rated voltage and 2/3 rated current applied to the meter (see 3.6.2).

4.6.8.1 It is the intent to specify a test coil which will produce a uniform field of approximately 5 oersteds through the case of the meter. Therefore, coils of other sizes are equally suitable if an equivalent electrical field is produced.

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4.6.9 Temperature influence.- Meters shall be tested at temperatures 20° C. above and below the reference temperature. Readings shall be taken at not less than four points on the scale, and the temperature influence shall be considered one-half the percentage change in indication, caused by the 20° C. departures from the reference temperature. If the influences above and below the reference temperature are not equal, the greater value shall be considered the temperature influence (see 3.6.2).

4.6.9.1 Heat effect at 65° C.- Meters shall be maintained at reference temperature for a sufficient time (not less than 2 hours) to attain thermal stability, and shall then be subjected to the three successive cycles of temperature change listed in table IV. During this test, meters shall indicate freely at all parts of the scale (see 3.6.2).

Table IV - Procedure for determining heat effect at 65° C.

Cycle number	Period (hours)	Step number	Procedure
-	--	1	Take initial reading at reference temperature.
-	16	2	Place meter in chamber at 65° C. and maintain that temperature for the remainder of that period.
-	--	3	Read meter while at 65° C. at the end of step 2.
-	8	4	Allow chamber and meter to cool down to the reference temperature during this period.
1	--	5	Read meter while at reference temperature at the end of step 4.
-	16	6	Raise temperature of chamber and meter as rapidly as possible to 65° C. and maintain that temperature for the remainder of the period.
2	--	7	Read meter while at 65° C. at the end of step 6.
-	8	8	Allow chamber and meter to cool down to the reference temperature during this period.
-	--	9	Read meter while at reference temperature at the end of step 8.
3	16	10	Raise temperature of chamber and meter as rapidly as possible to 65° C. and maintain that temperature for the remainder of the period.
-	--	11	Read meter while at 65° C. at the end of step 10.
3	8	12	Allow chamber and meter to cool down to the reference temperature during this period.
-	--	13	Read meter while at reference temperature at the end of step 12.

4.6.9.2 Low-temperature exposure.- Meters shall be exposed to a temperature of 0° C. for 16 hours, and at the end of the period meter readings shall be taken. It shall be observed whether or not the meter indicated freely at all parts of the scale while at 0° C. Meters shall then be maintained at room temperautre for sufficient time to attain thermal stability, and meter readings repeated. The permanent change in indication resulting from the low-temperature exposure shall be noted (see 3.6.2)

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4.6.10 Overload.-

4.6.10.1 Momentary overload.- The current circuits of a.c. and d.c. meters shall be subjected to 10 applications of current equal to 10 times end-scale value. The current shall be applied nine times for a period of one-half second each time, with 1-minute intervals between successive applications, then followed by one application for a 5-second period (making a total of 10 applications during an elapsed period of 9 minutes, 8-1/2 seconds), during which time current is applied for a total of 8-1/2 seconds. Normal voltage shall be applied to the meter during the whole period of the test. After 1 hour at reference temperature following the above test, the pointer shall be reset and the change in indication determined. This test should not be applied to external shunts (see 3.6.2).

4.6.10.2 Sustained overload.- Meters shall be subjected for 8 hours to an application of energy 20 percent greater than end-scale value. Immediately after the load is withdrawn at the end of the 8-hour period the temporary shift shall be determined. Sixteen hours after completion of the 8-hour overload period, the permanent shift shall be noted.

4.6.11 Voltage influence.- The power factor meter shall be tested to determine the effect of a plus and minus voltage variation from rated voltage.

4.6.12 Dielectric stress.- When performing this test, portable meters with nonmetallic cases shall be placed on a metallic platform; switchboard meters with nonmetallic cases shall be mounted on a metallic panel. All meters shall be subjected to a voltage of 2,600 volts r.m.s. applied between all external live parts of the meter and:

(a) The metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic.
(b) The zero-adjuster; and range selector switch, when applicable.
(c) All exposed metal parts not a live part of the meter.

The voltage shall be nominal 60 cycles per second in frequency and shall approximate, as closely as possible, a true sine wave in form. The voltage shall be applied continuously for a period of 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer, or by means of an auxiliary ratio transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually, and upon completion of the 60-second test period shall be gradually reduced. The test shall be made using high-potential test prods having not less than 1/8-inch radius tips. Any damage or flash-over shall be noted (see 3.6.2).

4.6.13 Insulation resistance.- Meters shall be placed on a metallic platform or mounted on a metallic panel as specified in 4.6.12. Using a potential of 1100 volts direct current, the insulation resistance shall be determined between all external live parts and the metallic platform or panel on which the meter is placed or mounted, or the meter case, if the case is metallic (see 3.6.2).

4.6.13.1 For inspection tests of meters furnished with external accessories as an integral part the complete meter (meter with accessories) shall be tested.

4.6.14 Vibration.- Power factor meters shall be subjected to the type I tests of Standard MIL-STD-167. The change in indication shall then be determined (see 3.6.2).

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4.6.15 Friction effect.- Upon completion of the vibration test, the friction effect shall be determined, as follows: Untapped ascending and descending readings shall be taken at five points distributed over the scale, using the same scale points for ascending and descending readings. The difference between corresponding ascending and descending readings as observed on the standard meter shall be determined (see 3.6.2).

4.6.16 Humidity resistance.- Meter readings shall be taken at reference temperature. The meter shall then be placed in a humidity chamber and kept for 6 hours at a temperature of 65° C. with relative humidity of 90 percent. Sufficient electrical energy shall be supplied to the meter to maintain two-thirds end-scale deflection. At the end of the 6-hour period, the heat shall be shut off and the meter de-energized. The meter shall then be permitted to cool for 18 hours in this atmosphere, the relative humidity of which rises as the temperature is reduced. (Precaution shall be taken to prevent dripping on the meter.) A temperature of 35° C. or lower, but not less than 25° C., shall be attained during the cooling period. The complete cycle shall then be repeated. Upon completion of the second cycle, the meter shall be removed from the humidity chamber and permitted to dry for a period of 24 hours at room temperature, with relative humidity less than 60 percent. Meter readings shall then be taken at reference temperature. The meter window shall then be removed and the meter, with window removed, subjected to two additional cycles. The meter shall then be removed from the humidity chamber and permitted to dry for a period of 24 hours at room temperature, with relative humidity less than 60 percent. Meter readings shall then be taken at reference temperature. After the drying period, a maximum of 10 momentary end-scale deflections will be permitted prior to the final reading. Only one meter of the sample shall be subjected to this test (see 3.6.2).

4.6.17 Shock (high-impact).- High-impact-shock-resistant meters shall be tested in accordance with type C of Specification MIL-S-901 for lightweight equipment, when mounted on a plate as shown on figure 6D of that specification. The meter of the sample, which was not subjected to the humidity-resistance test, shall be used. During the test, the meter shall be energized to approximately one-half scale deflection A total of nine blows shall be applied. Three blows shall be applied parallel to each of the three principal axes of the meters, the three blows for each direction to be with heights of hammer drop 1 foot, 3 feet, and 5 feet. The pointer shall be reset to zero after each blow. At the completion of the shock test, the meter shall be checked for calibration errors, then disassembled and carefully examined for damage or misalignment of parts (see 3.6.2).

4.7 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 For domestic shipment - immediate use.- Packaging shall be in accordance with the manufacturer's commercial practice.

5.1.2 For domestic shipment and storage or overseas shipment.- Each power factor meter shall be individually cushioned and unit packaged in accordance with method IIb of Specification MIL-P-116

5.2 Packing.-

5.2.1 For domestic shipment - immediate use.- Power factor meters packaged as specified in 5.1.1 shall be packed in a manner to insure acceptance and safe delivery at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

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5.1.1 For domestic shipment and storage.- Power factor meters packaged as specified in 5.1.1 shall be packed in wood-cleated fiberboard, wood-cleated plywood, wood-kraft veneer board, nailed wood, or fiberboard boxes conforming to Specifications NN-B-591, PPP-B-601 (domestic type MIL-B-10377 (domestic type), NN-B-621, or the special requirements of Specification LLL-B-631 or LLL-B-636. Boxes shall be closed in accordance with the applicable box specification. The gross weight of wood or wood-cleated containers shall not exceed 200 pounds, of fiberboard containers, 70 pounds.

5.1.2 For overseas shipment.- Power factor meters packaged as specified in 5.1.2 shall be packed in wood-cleated fiberboard, wood-cleated plywood, wood-cleated kraft-veneer board, nailed wood or fiberboard boxes conforming to Specification JAN-P-108, PPP-B-601 (overseas type), MIL-B-10377 (overseas type), JAN-P-106, or JAN-P-108 (exterior grade). Boxes shall be sealed, closed, and strapped in accordance with the appendix of the applicable box specification. Wood or wood-cleated containers shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds, of fiberboard boxes, 70 pounds.

5.3 Marking.- Unit packages shall be marked in accordance with Standard MIL-STD-129 with the following information:

Stock number.
Nomenclature.
Quantity.
Contract or order number.
Manufacturer's name and address.

In addition to any special marking required by the contract or order, shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Power factor meters specified in this specification are primarily for use in switchboard applications where the power factor of the circuit is required to be known. Power factor meters may also be used for such application as salinity indicators where the basic principles of a power factor meter is useful in calibration to other than electrical quantities. When this is the case the requirements of the basic equipment (such as a salinity indicator) specification have precedence over the requirements of this specification.

6.2 Ordering data.- Procurement documents should specify the following

a) Title, number, and date of this specification.
b) Complete rating information including whether the power factor meter shall be single or polyphase.
c) Adapters, if required.
d) External accessories, if required (see 3.5.10).
e) Whether packaging and packing are for domestic shipment - immediate use domestic shipment and storage or overseas shipment (see 5.1 and 5.2).

6.3 Indirect shipments.- The packaging, packing, and marking requirements specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and prime contractor.

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6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in Qualified Products List QPL-19261 whether or not such products have actually been so listed by that date. Manufacturers are urged to communicate with the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C. and arrange to have the products that they propose to offer tested or qualification.

6.4 Matters relating to the use of power factor meters furnished in accordance with this specification for use in the design, production, and procurement of equipment should be referred to that department or service whose equipment is involved.

6.5 Precautions.- Attention of equipment designers and manufacturers is invited to the fact that sufficient space should be provided in equipment for mounting standard meters of the maximum depth specified herein even though a shallower meter may be in the initial design.

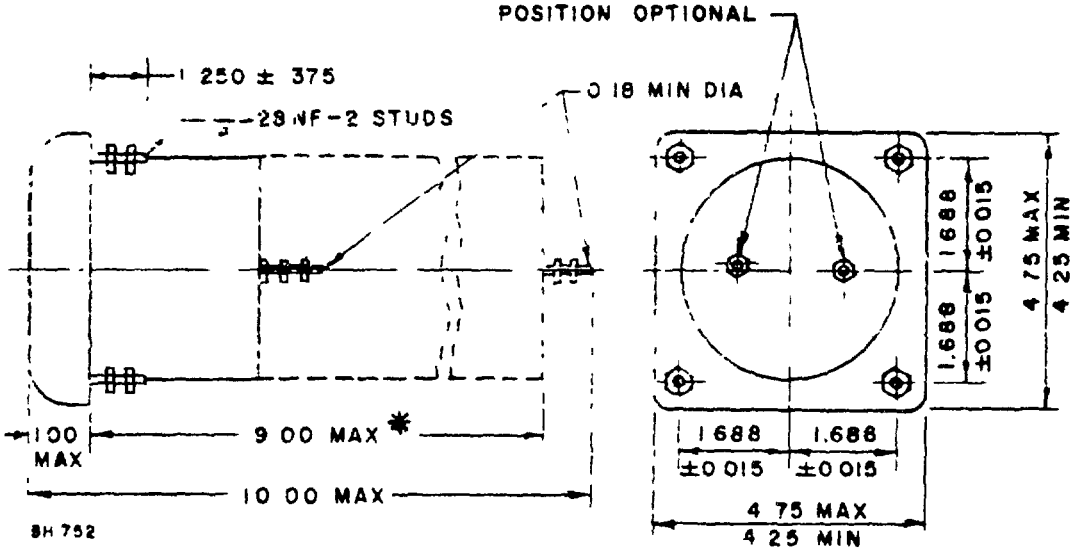
6.6 This specification replaces that part of Specification 17-M-25 covering power factor meters.

6.7 Specification MIL-W-19088 replaces that part of Specification 17-M-25 covering wattmeters.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs not responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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* THIS DIMENSION HAS BEEN CHOSEN TO ALLOW THE MOUNTING OF ACCESSORIES ON INSTRUMENT PROPER

ALL DIMENSIONS IN INCHES.

Figure 1 - Dimensions and drilling plans for power factor meter 4-1/2 in, switchboard type.

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